

MSAS – Assignment #1: Simulation

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1 Implicit equations

Exercise 1

Consider the physical model of a mechanical system made of four rigid rods $\mathbf{a}_i$ with $i \in [1; 4]$ in a kinematic chain, as shown in Fig. 1. The system has only two degrees of freedom. The objective is to determine, fixing the angle β , the corresponding angle of equilibrium α between the rods $\mathbf{a}_1$ and $\mathbf{a}_2$. Writing the kinematic closure, it is possible to obtain the following equation

$$\frac{a_1}{a_2} \cos \beta - \frac{a_1}{a_4} \cos \alpha - \cos(\beta - \alpha) = -\frac{a_1^2 + a_2^2 - a_3^2 + a_4^2}{2a_2a_4} \quad (1)$$

where a_i is the length of the corresponding rod. This equation, called the equation of Freudenstein, can be clearly re-written in the form $f(\alpha) = 0$ and solved, once β is fixed, for a particular value of α . Assume that $a_1 = 10$ cm, $a_2 = 13$ cm, $a_3 = 8$ cm, and $a_4 = 10$ cm and

- 1) Implement a general-purpose Newton solver (NS).
- 2) Solve Eq. (1) with NS for $\beta \in [0, \frac{2}{3}\pi]$ with the analytical derivative of $f(\alpha)$ and tolerance set to 10^{-5} . For every β value, use two initial guesses, -0.1 and $\frac{2}{3}\pi$. What do you observe? You can validate your results with the Matlab[®] built-in function `fzero`.
- 3) Repeat point 2) using the derivative estimated through finite differences and compare the accuracy of the two methods.
- 4) Eq. (1) can be solved analytically only for specific values of β . Find at least one and compare the analytical solution with yours.
- 5) Repeat point 2) extending the β interval up to π . What do you observe?

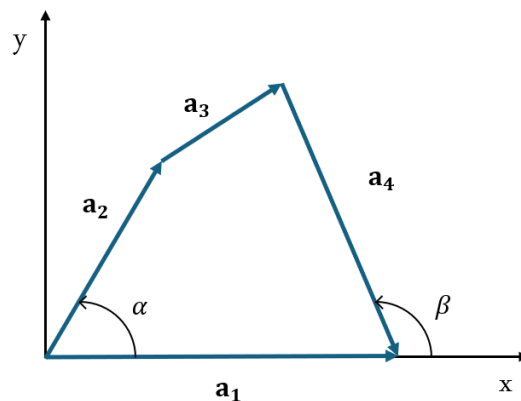
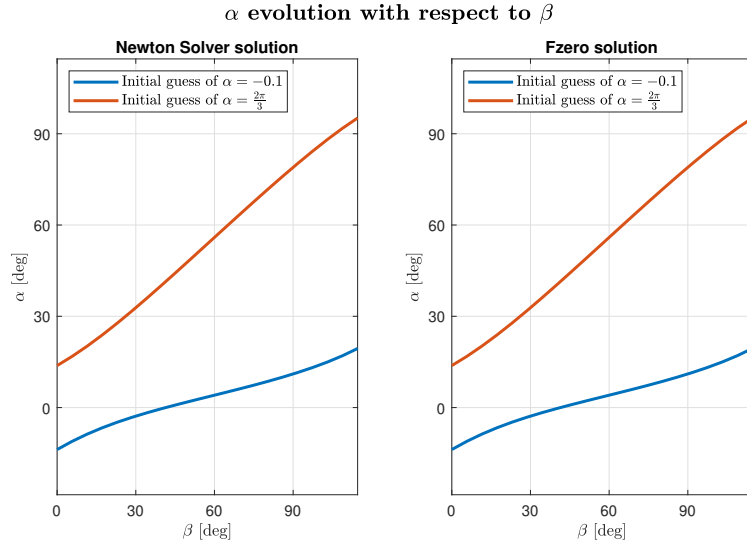
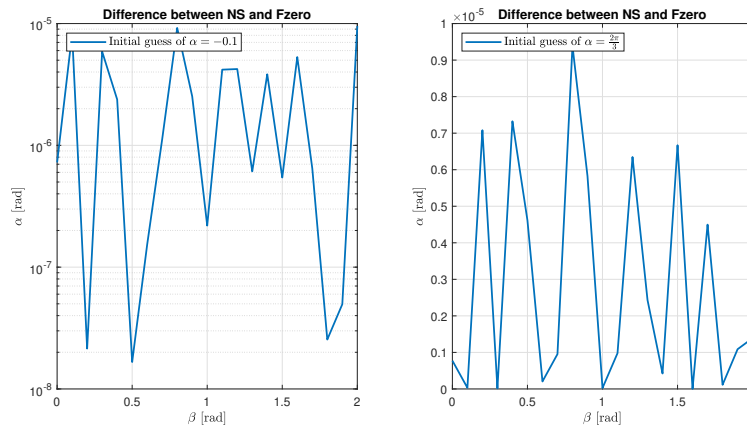


Figure 1: Kinematic chain.

(4 points)

Solution

1) Newton Solver(NS) was implemented to iteratively solve $f(\alpha) = 0$, as defined in Eq. 1. The function derivative was obtained from the `syms` function of *Matlab*, which was implemented

**Figure 2:** Solutions obtained from Newton solver and *fzero***Figure 3:** Error of Newton solver with respect to the *fzero*

for the unknown α . It was considered an iteration limit of 50, such that the solver doesn't become computationally expensive.

$$\alpha(k+1) = \alpha - \frac{f(\alpha)}{df(\alpha)/d\alpha} \quad (2)$$

The process is continued until the desired tolerance is achieved.

2) The tolerance was set to 10^{-5} for the NS. In order to check the results, the *Matlab*-integrated function, *fzero*, has been used. The difference between the two methods are shown in Fig. 3. Fig. 2 shows the obtained results for the two initial guesses $\alpha_1 = -0.1$ and $\alpha_2 = \frac{2}{3}\pi$. As it is noticed, the two results vary with respect to the initial values for both the NS and the *fzero*, offering thus two possible α angles for each β . This behaviour is understandable, as the Freudenstein equation is non-linear and thus for each value of β more than one value of α could be found in a limited range of angles. It is interesting to note the strong dependence of both methods on the initial conditions. It is clear that for an initial value of $\alpha = \frac{2}{3}\pi$ the Newton Solver marginally outperforms *fzero*, as can be seen in Fig. 3. NS offers a good solution, considering the tolerance of 10^{-5} .

3) The process of implementing NS was similar, the difference consisting of how $f'(\alpha)$ was calculated. In this case, $f'(\alpha)$ was calculated using the finite difference method, and Finite

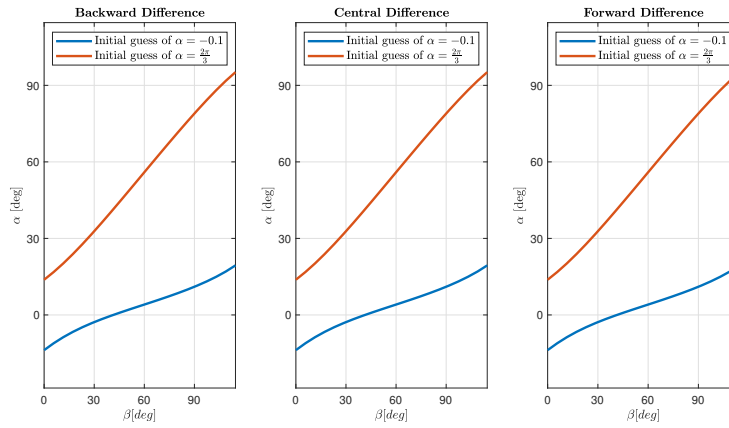


Figure 4: Finite Differences methods.

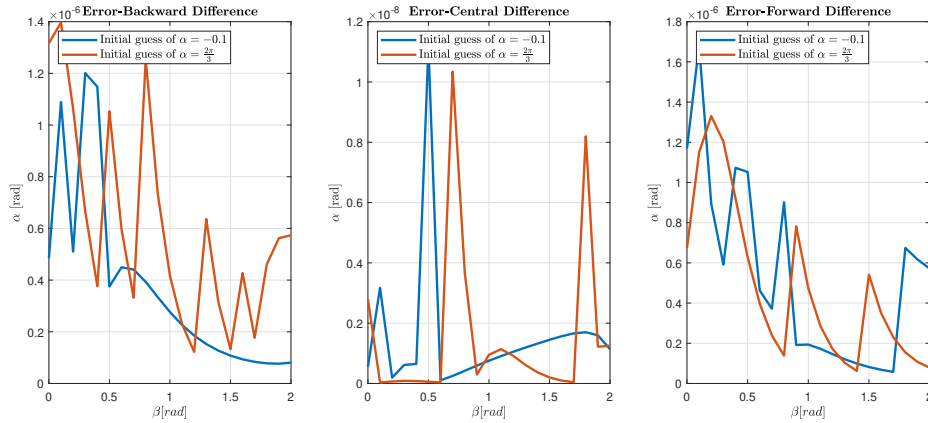


Figure 5: Error between Newton Solver and Finite Differences

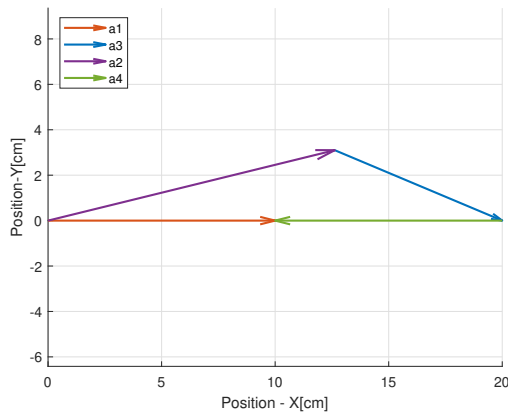
difference methods (Backward, Central, Forward) were implemented for derivative estimation. Fig. 4 shows the results for each method, which closely match those of the analytical derivative. As shown in Fig. 5, the Central Difference method provides the smallest error, while Forward and Backward differences also meet the tolerance of 10^{-5} . Central Difference is recommended for applications requiring higher precision.

4) For comparison, the analytical method was also calculated. In order to solve the problem analytically, two cases were considered which were solved using the *Cosine Theorem*.

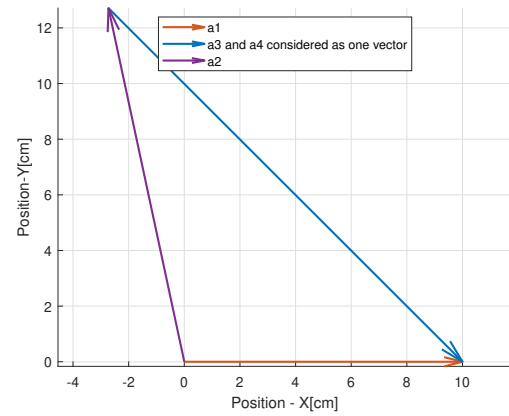
- The vectors a_3 and a_4 have the same direction. For this case it was obtained that $\alpha = 102.21$ and $\beta = 135.1$ (Fig. 6a).
- The vectors a_1 and a_4 have the same direction along the x-axis but opposite orientations. In this case it is obvious that it was considered that $\beta = 0$ and $\alpha = 13.7953$ (Fig. 6b).

Both obtained solutions correspond to the solutions obtained numerically, through NS.

5) If the interval for β is extended up to π , the solution diverges, accumulating errors. This is due to the limits imposed by the constrained vectors. Thus, for values of β larger than 150° , the system cannot move anymore (it becomes locked). When a solver tries to continue beyond this boundary, the mathematical model can yield unrealistic results, often resulting in extremely large values due to the inability of the solver (both *fzero* and NS) to interpret the physical constraints correctly. Beyond the boundary, the nonlinear terms in the Freudenstein equation can become exponentially sensitive to small changes in β . The result leads to extremely large values that are mathematically consistent with the equation, but not physically meaningful.

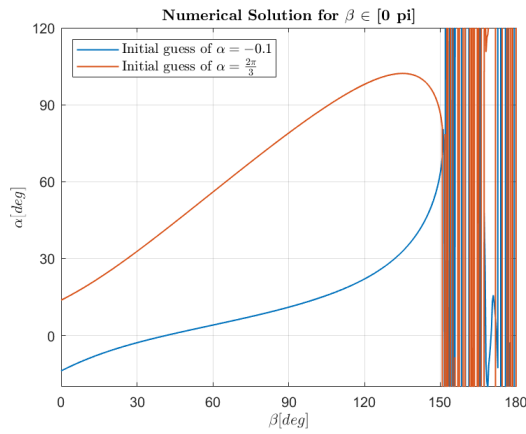


(a) First Analytical Case considered

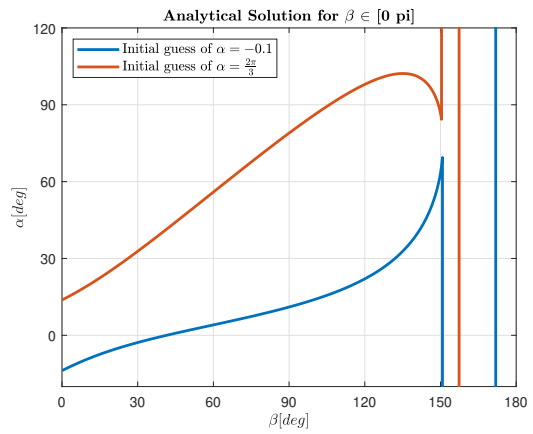


(b) Second Analytical Case considered

Figure 6: Analytical Solutions



(a) Newton Solver solution



(b) *fzero* solution

Figure 7: Solutions for $\beta \in [0 \pi]$

Fig. 7. Thus the solver goes on until it reaches π , where it stop as the value obtained is ∞ ($f(\alpha) = \text{const}$).

2 Numerical solution of ODE

Exercise 2

The physical model of a moving object with decreasing mass is shown in Fig. 8. The mathematical model of the system is expressed by the equations:

$$m(t) \frac{dv}{dt} = \bar{F} + f(t) - \alpha \cdot v \quad (3)$$

$$m(t) = m_0 - c_m \cdot t \quad (4)$$

$$\bar{F} = -0.5 \cdot \rho \cdot C_d \cdot A_m \cdot v^2 \quad (5)$$

Assuming that: $v(0) = 0$ [m/s], $m_0 = 20$ [kg], $c_m = 0.1$ [kg/s], $f(t) = 1$ [N], $\alpha = 0.01$ [Ns/m], $\rho = 0$ [kg/m³], and Eq. (3) has the exact solution:

$$v(t) = \frac{f(t)}{\alpha} - \left[\frac{f(t)}{\alpha} - v(0) \right] \left[1 - \frac{c_m \cdot t}{m_0} \right]^{\frac{\alpha}{c_m}} \quad (6)$$

Answer to the following tasks:

- 1) Implement a general-purpose, fixed-step Heun's method (RK2);
- 2) Solve Eq. (3) using RK2 in $t \in [0, 160]$ s for $h_1 = 50$, $h_2 = 20$, $h_3 = 10$, $h_4 = 1$ and compare the numerical vs the analytical solution;
- 3) Repeat points 1)–2) with RK4;
- 4) Trade off between CPU time & integration error.

Now, assuming that: the fluid density is $\rho = 900$ [kg/m³], $C_d = 2.05$, and $A_m = 1$ [m²], answer to the following tasks:

- 1) Solve Eq. (3) using RK2 in $t \in [0, 160]$ s for $h_1 = 1$;
- 2) From the results of the previous point, use a proper ode of Matlab to solve Eq. (3) ;
- 3) Discuss the results.

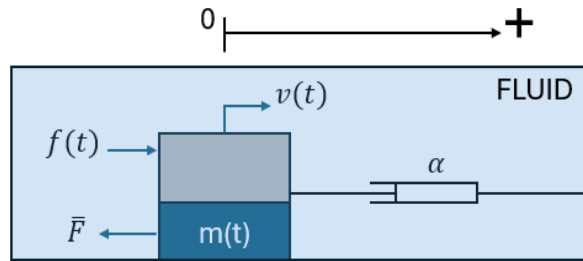


Figure 8: Moving object with decreasing mass.

(5 points)

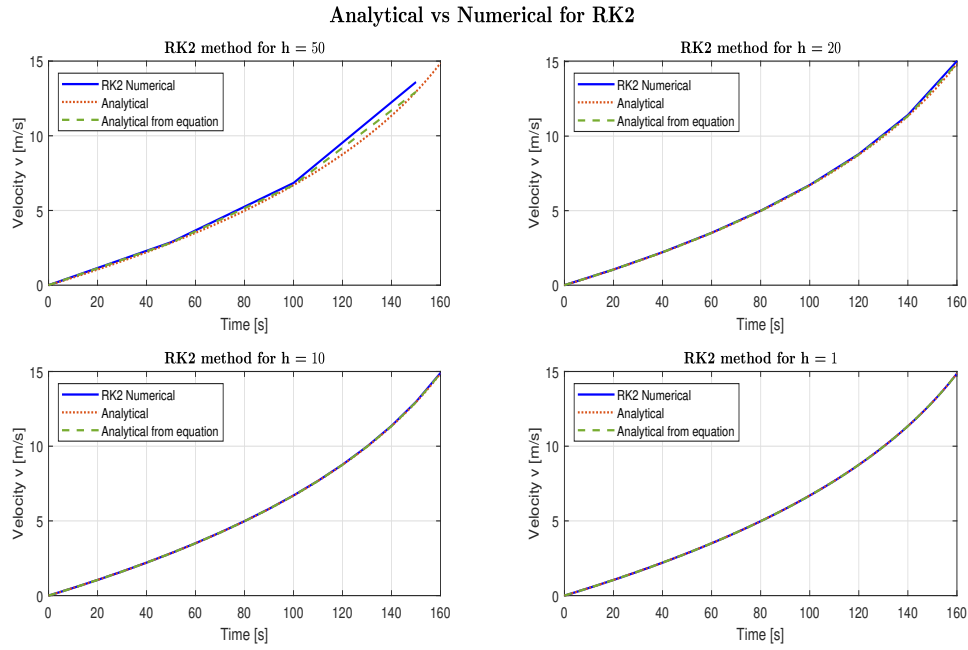


Figure 9: Heun's method (RK2) compared to analytical method for each step size h

Solution

Part I

1) The Heun's method (RK2) was implemented using fixed-step integration with coefficients α and β defined as:

$$\alpha = [1] \quad \beta = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (7)$$

2) For the RK2 the solution was implemented considering the above mentioned Heun's method, for the given size steps, in the given interval. The procedure includes an iteration over the time interval, for a set number of steps $n(h)$. The number of steps is calculated with the Matlab function $\text{round}(\frac{\Delta t}{h})$, as it is not wanted to exceed the interval's bounds. The velocity is calculated at fixed time steps, that are updated as follows:

$$t_{k+1} = t_k + h \quad (8)$$

Thus for the first step size, RK2 cannot cover the entire interval, as it reaches the last step at $t = 150$ s, owing to large step size, while all the other steps sizes manage to cover the entire interval. For this case, the analytical solution was calculated using both the equation of the exact solution that is dependent on time, and thus the velocity was calculated at each exact time point obtained from the h , and by using the integrated *ode45* integrator. The results are shown in Fig. 9. It is noticeable that at small step sizes the error is larger while, as step size decreases, the accuracy of the RK2 method improves, which can also be observed in Fig. 11.

3) The implementation of RK4 was similar to that of RK2, considering the coefficients from Eq. 9, for each time step, which has been calculated using Eq. 8.

$$\alpha = \begin{bmatrix} 1 \\ \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \quad \beta = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{bmatrix} \quad (9)$$

The results are presented in Fig. 11, with respect to the analytical solution as done in the previous case.

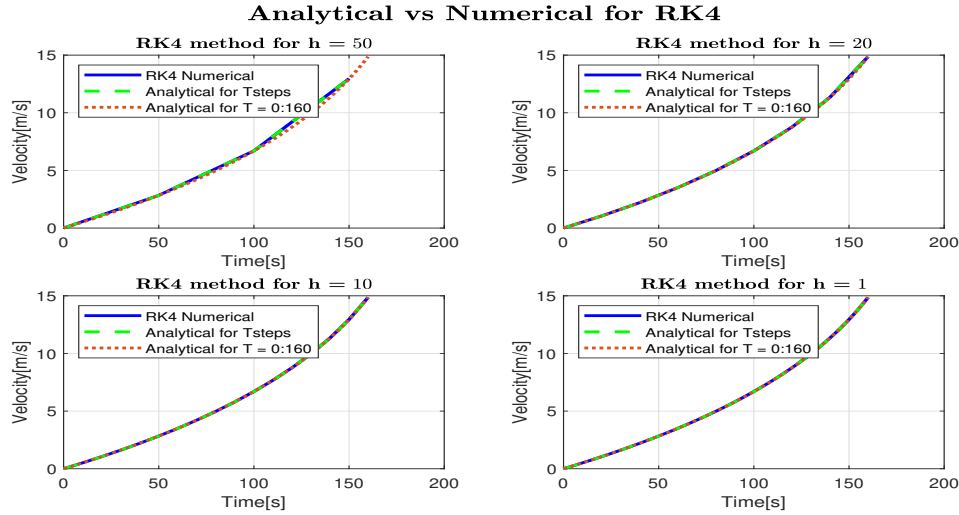


Figure 10: RK4 method compared to Analytical Solution for each step size h

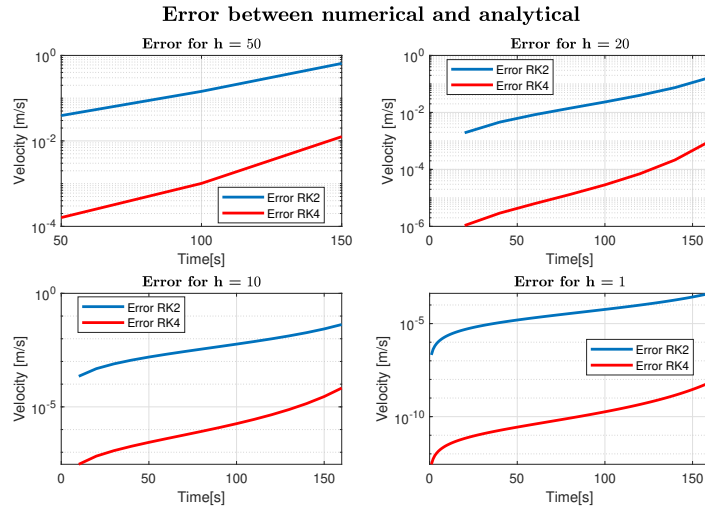


Figure 11: Errors of RK2 and RK4 with respect to analytical solution for each step size h

4) In Fig. 10, the error for both RK2 and RK4 are plotted with respect to the analytical solution, while in Fig. 12 is shown the difference in performance between RK2 and RK4 for each step size. It is clear that for this system the RK4 outperforms RK2, offering a much smaller error, as expected. However, regarding the computational efficiency needed by each method the RK2 method delivers faster results completing the process in almost 0.79s, while it takes RK4 around 1.34s to complete the calculations.

Part II

1 - 2 - 3) As now the density is considered, $\rho = 900$, also the force $\bar{F}$ included is the system, which becomes:

$$\frac{dv}{dt} = -0.5 \cdot \rho \cdot C_d \cdot \frac{A_m}{m(t)} \cdot v^2 + \frac{1}{m(t)} \cdot (m_0 - c_m \cdot t) - \alpha \cdot v \quad (10)$$

Eq. 10 is implemented to a RK2 method, similar to the one already implemented in the previous cases, considering the given step size. Similarly, an *ode45* was implemented. The integrator required time interval was the one provided.

However, the RK2 model fails to complete the calculations for the given step size, $h = 1$ being too large for this case, as it can be seen in Fig. 13. If h is decreased, the error decreases

until it reaches a similar solution to the analytical one. From a theoretical standpoint, a larger step size would trade the accuracy for the computational efficiency. However, in this case as h is large enough, such that the error accumulates reaching values too large for *Matlab* to continue the computation and thus, also the efficiency is sacrificed. Therefore, a smaller step size is desired from both efficiency and accuracy point of view.

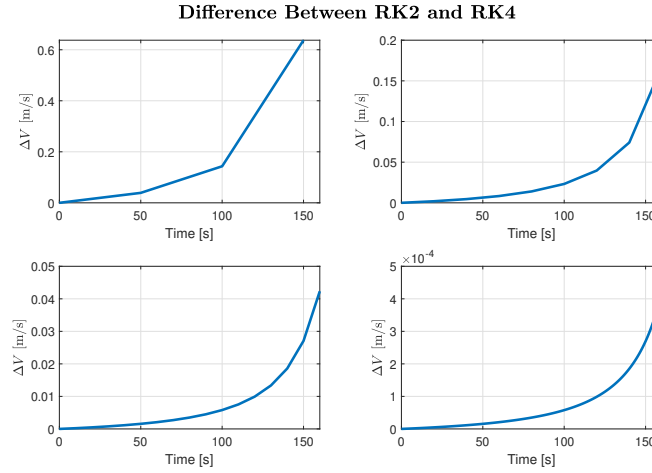


Figure 12: Simulation error of RK2 with respect to RK4

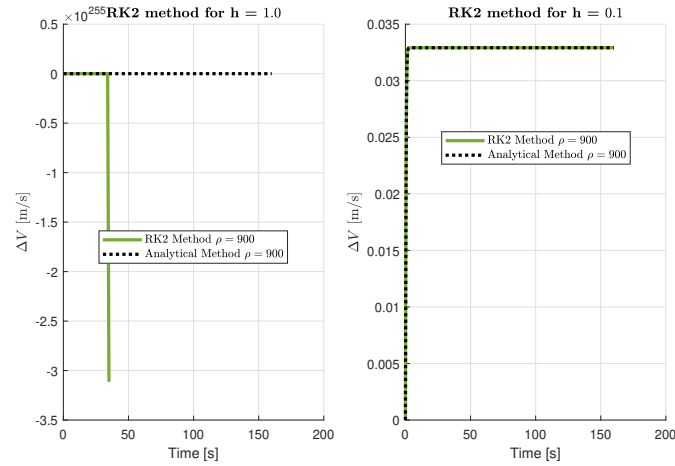


Figure 13: Simulation for $\rho = 900$ for $h = 1$ and $h = 0.1$

Exercise 3

Let $\dot{\mathbf{x}} = A(\alpha)\mathbf{x}$ be a two-dimensional system with $A(\alpha) = [0, 1; -1, 2 \cos \alpha]$. Notice that $A(\alpha)$ has a pair of complex conjugate eigenvalues on the unit circle; α denotes the angle from the $\text{Re}\{\lambda\}$ -axis.

a) Runge-Kutta methods

1) Write the operator $F_{\text{RK2}}(h, \alpha)$ that maps $\mathbf{x}_k$ into $\mathbf{x}_{k+1}$, namely $\mathbf{x}_{k+1} = F_{\text{RK2}}(h, \alpha) \mathbf{x}_k$. 2) With $\alpha = \pi$, solve the problem “Find $h \geq 0$ s.t. $\max(|\text{eig}(F(h, \alpha))|) = 1$ ”. 3) Repeat point 2) for $\alpha \in [0, \pi]$ and draw the solutions in the $(h\lambda)$ -plane. 4) Repeat points 1)–3) with RK4.

b) Backinterpolation methods

Consider the backinterpolation method $\text{BI}_{2,0.3}$. 1) Derive the expression of the linear operator $B_{\text{BI}_{2,0.3}}(h, \alpha)$ such that $\mathbf{x}_{k+1} = B_{\text{BI}_{2,0.3}}(h, \alpha)\mathbf{x}_k$. 2) Using the same approach of a), draw the stability domain of $\text{BI}_{2,0.3}$ in the $(h\lambda)$ -plane. 3) Derive the domain of numerical stability of $\text{BI}_{2,\theta}$ for the values of $\theta = [0.2, 0.4, 0.6, 0.8]$.

(8 points)

Solution

a) Runge - Kutta methods

1) Starting from the RK2 system of equations and the TSE, the predictor was introduced into the corrector and it was accounted that:

$$f(x) = \dot{x} = A(\alpha) \cdot x \quad (11)$$

Thus, for the $F_{\text{RK2}}(h, \alpha)$ was obtained the following formula:

$$F_{\text{RK2}}(h, \alpha) = \mathbf{I}_2 + h \cdot \mathbf{A} + \frac{h^2}{2} \cdot \mathbf{A}^2 \quad (12)$$

which if introduced in the formula of x_{k+1} one obtains:

$$x_{k+1} = F_{\text{RK2}}(h, \alpha)x_k \quad (13)$$

2) The solution of h for $\alpha = \pi$ was obtained by applying *fsolve* to the given constraint. The tolerance was considered the default one (10^{-6}). As *fsolve* is sensitive to the initial conditions, the initial conditions was consider to belong to the interval $[1 \ 2]$, as one of the solution that satisfies the equation is zero, and thus if the initial condition of the step size were to be considered smaller than 1, the problem would result in a column vector with values close to zero. Finally, the obtained result is $\mathbf{h} = \mathbf{2}$, for $h_0 = 2$.

3) Continuing from the previous point, the initial value was considered $h_0 = 2$, for $\alpha = \pi$. For the subsequent values of α , until it reached 0, the initial values were taken from numerical continuation, considering the result of the previous step. The result is shown in Fig. 14.

4) Similar to the case of RK2, the RK4 was implemented considering F_{RK4} operator as:

$$F_{\text{RK4}}(h, \alpha) = \mathbf{I}_2 + h \cdot \mathbf{A} + \frac{h^2}{2} \cdot \mathbf{A}^2 + \frac{h^3}{6} \cdot \mathbf{A}^3 + \frac{h^4}{24} \cdot \mathbf{A}^4 \quad (14)$$

From this formula was calculated the solution of step size for $\alpha = \pi$ to be equal with $h = 2.7853$, which was obtained considering an initial value $h_0 = 2$, as it was observed that around the value of 2 are the second solutions of the equation. Similarly to RK2, it was used numerical continuation to redefine the initial condition for each α values, to continue on the interval $[0 \ \pi]$.

When the solution of the step size reached a value of 2.2, it was observed a nonlinearity in the solutions given by *fsolve*, Fig. 15b. The RK4 stability area was obtained by finding the

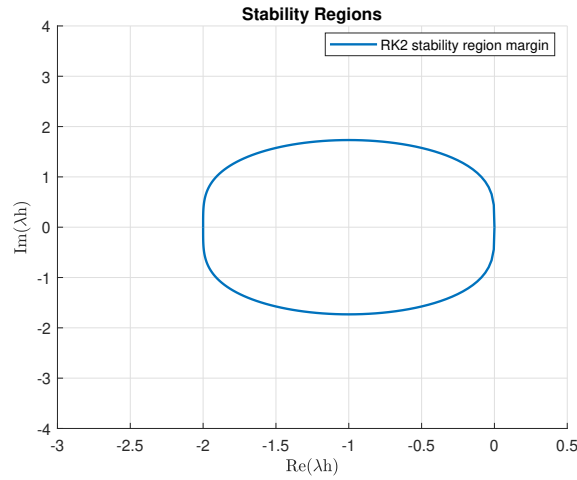


Figure 14: RK2 stability Region

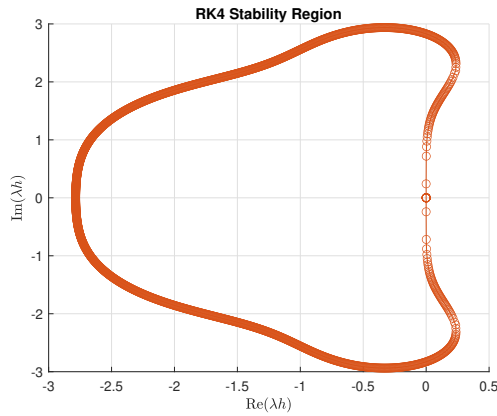
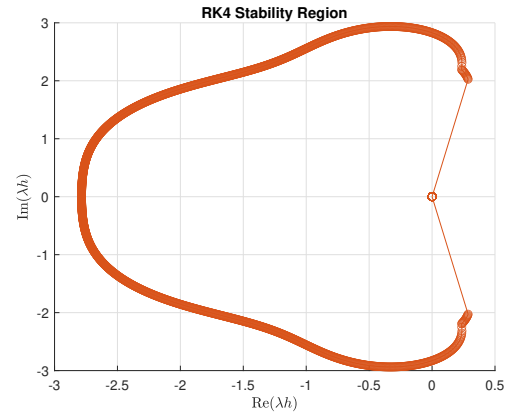

 (a) RK4 stability area stopping the value of $\alpha = 1.46$

 (b) RK4 stability area for $\alpha \in [0 \pi]$

Figure 15: RK4 Stability area

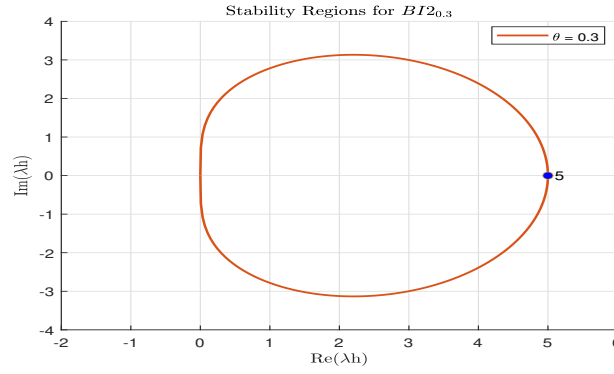
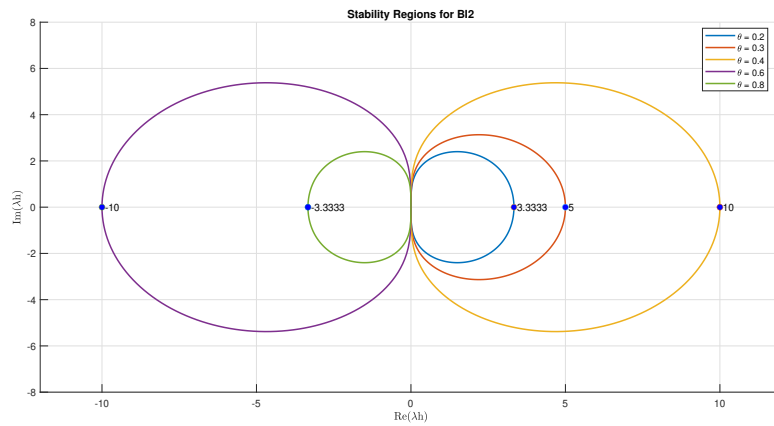
solutions for step size with respect to α until it reached $h = 2.2$. When this value was reached, the numerical continuity process was continued, however the value of α was reversed back to π . When the value of h reaches zero the process was stopped. The result is shown in Fig. 15a. To plot the results in the $h\lambda$ -plane, the solutions obtained for h were multiplied by the eigenvalues of $\mathbf{A}$ for each value of α . RK4 achieves a larger stability region compared to RK2, as seen in Fig. 15b.

b) Backinterpolation methods

$$F_{TR} = [\mathbf{I}^{(n)} - \mathbf{A} \cdot \frac{h}{2}]^{-1} \cdot [\mathbf{I}^{(n)} + \mathbf{A} \cdot \frac{h}{2}] \quad (15)$$

1) Similar to the previous two cases the $BI_{2,0.3}$ was obtained starting from general BI algorithms of 1st order, the trapezoidal rule, to which the θ multiplier was applied. By definition, the semi-step of trapezoidal rule θ is considered to be 0.5, known as the implicit midpoint rule. θ shall be considered as the multiplier that settles the fraction of steps with the explicit method. For the operator $BI_{2,0.3}$, $\theta = 0.3$. Thus it can be written:

$$B_{BI_{2\theta}} = \left(\mathbf{I}_2 - (1 - \theta) \cdot h \cdot \mathbf{A} + \frac{(\mathbf{A} \cdot h \cdot (1 - \theta))^2}{2} \right)^{-1} \left(\mathbf{I}_2 + \theta \cdot h \cdot \mathbf{A} + \frac{(\mathbf{A} \cdot \theta \cdot h)^2}{2} \right) \quad (16)$$

**Figure 16:** BI stability region for $\theta = 0.3$ **Figure 17:** BI stability regions for θ

$$B_{BI2_{0.3}} = \left(\mathbf{I}_2 - 0.7 \cdot h \cdot \mathbf{A} + \frac{(\mathbf{A} \cdot h \cdot 0.7)^2}{2} \right)^{-1} \left(\mathbf{I}_2 + 0.3 \cdot h \cdot \mathbf{A} + \frac{(\mathbf{A} \cdot 0.3 \cdot h)^2}{2} \right) \quad (17)$$

2) In order to draw the stability domain of BI, a set of initial conditions were considered. The one that was the most suitable for this case was $h_0 = 5$, and for this was afterwards calculated the values of h using numerical continuation. The initial value was obtained from trial and error, several values considered larger or smaller than 5, were resulting in solutions that were not converging at the $\alpha = \pi$. The solution obtained is shown in Fig. 16.

3) To calculate for a range of θ values, Eq. 16 was implemented for each value of θ . The result for all values are shown in Fig. 17. As seen for $\theta = 0.3$, the value of initial state has a noticeable impact on the results, which is understandable as solutions can be found due to the based on the selected tolerance. For this reason, there were considered values that would converge for the considered tolerance, which are shown in Table. 1.

θ	0.2	0.3	0.4	0.6	0.8
h_0	10	5	10	-10	-10

Table 1: Initial Conditions

It can be noted the symmetry around the y-axis, that depends on the symmetry of θ around the value of 0.5, which is expected as it represents the symmetry around the midpoint where the Backward and Forward Interpolations have the same semi-step value, and thus carry the same importance.

Exercise 4

The cooling problem of a high-temperature mass is shown in Fig. 18. Assuming that: $K_c = 0.0042$ [J/(s K)] and $K_r = 6.15 \times 10^{-11}$ [J/(s K⁴)] are the convective and radiation heat loss coefficients respectively, $\tilde{T}$ is the mass temperature in time, $T_a = 277$ [K] is the surrounding air temperature, $C = 45$ [J/K] is the mass thermal capacity, and $\tilde{T}(0) = 555$ [K] is the initial mass temperature.

- 1) Write the mathematical model of the system.
- 2) Solve the mathematical model using: RK2 with $h = 720$, and RK4 with $h = 1440$.
- 3) Compare the solution of point 2) with the "exact" one. Discuss the results. (Hint: use an ODE solver of Matlab).

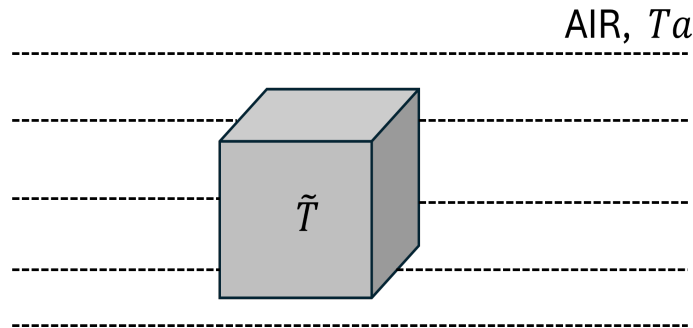


Figure 18: The cooling of high-temperature mass.

(4 points)

Solution

1) The body is affected by convection and radiation, thus the system can be represented by the following mathematical model:

1. Convective part:

$$Q_{conv} = -K_c \cdot (\tilde{T} - T_a) \quad (18)$$

2. Radiative part:

$$Q_{rad} = -K_r \cdot (\tilde{T}^4 - T_a^4) \quad (19)$$

3. Total Heat loss rate

$$Q_{Total} = Q_{conv} + Q_{rad} = C \cdot \frac{d\tilde{T}}{dt} \quad (20)$$

4. Differential equation describing the system

$$\frac{d\tilde{T}}{dt} = -\frac{K_c}{C} \cdot (\tilde{T} - T_a) - \frac{K_r}{C} \cdot (\tilde{T}^4 - T_a^4) \quad (21)$$

2 - 3) In order to solve the model with the RK2 and RK4 methods, it was considered that simulation should end when the model is in equilibrium. Thus, it was chosen a tolerance of 10^{-5} , such that the body should reach a temperature almost equal to that of the air surrounding it. It was assumed that the air around the body will have a constant temperature, thus the body being surrounded by a large amount of air, making the heat absorbed from the body irrelevant to the internal equilibrium of the mass of air. The results are shown in Fig. 19 and 20. It is observed that at the beginning the error for both methods is large and afterwards they start to

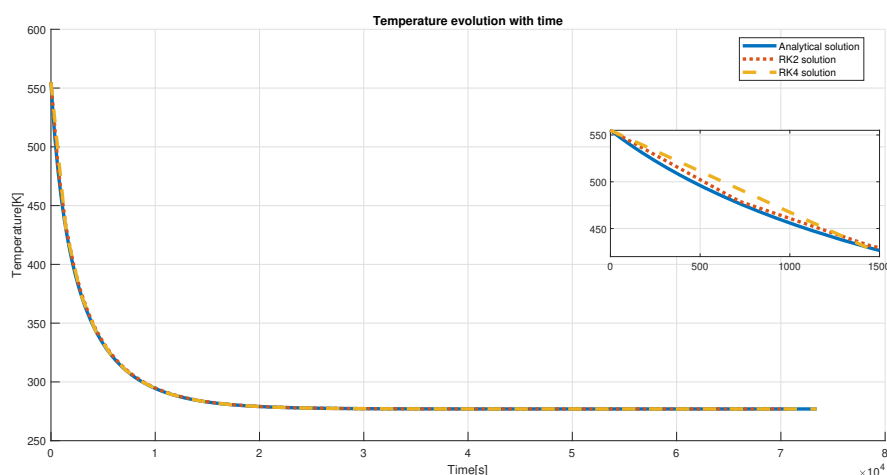


Figure 19: RK2, RK4 and analytical solution

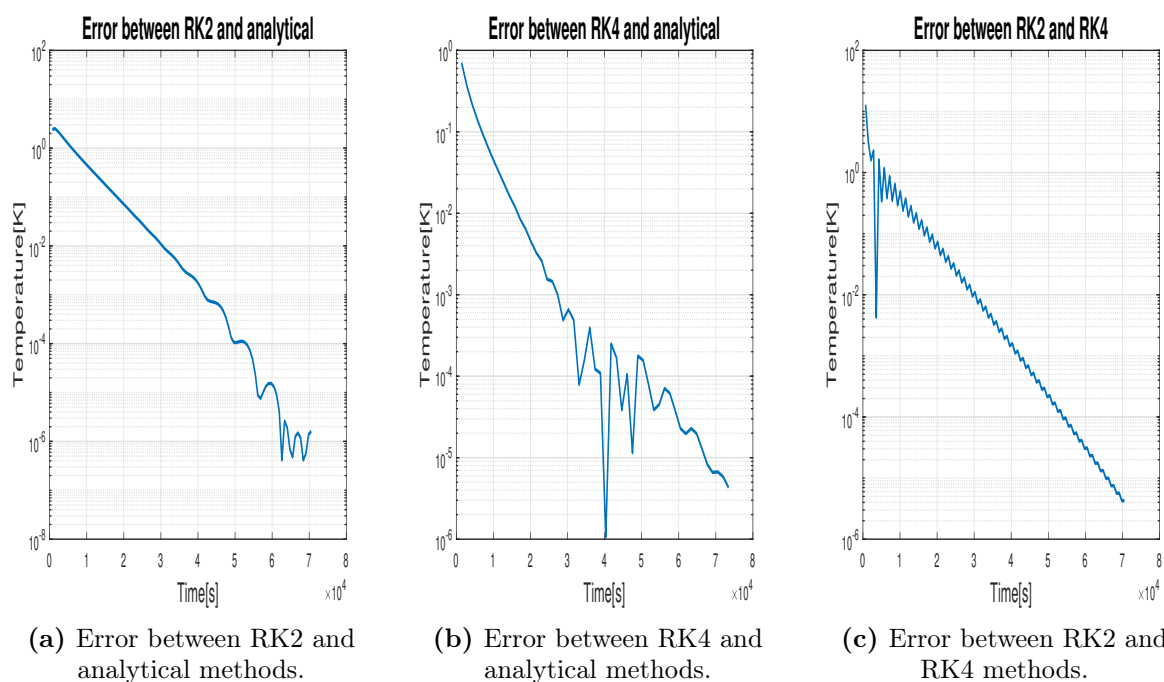


Figure 20: Error between RK2, RK4 and analytical methods

decrease, converging towards the analytical solution. From Fig. 20 can be observed that the error for the RK2 with respect to the analytical one is larger at the beginning than that of RK4 method, however, at the end the RK2 offers a smaller error and also it converges in a shorter period of time. This is due to the smaller step size considered for the RK2 method, as when the simulation was done with the same step size, the RK4 method had a faster rate of convergence.

Exercise 5

Consider the electrical circuit in Fig. 21. At time $t \geq 0$ the switch disconnects the battery and the capacitor discharges its stored energy to the circuit. Answer to the following tasks:

- 1) Find the state variable form of the circuit with $x_1 = q$ and $x_2 = \frac{dq}{dt}$, where q is the charge on capacitor.
- 2) Show that the system is stiff when the circuit parameters are $R = 25 \text{ } [\Omega]$, $L = 20 \text{ } [mH]$, $C = 200 \text{ } [mF]$, and $v_0 = 12 \text{ } [V]$. Represent the eigenvalues on the $(h\lambda)$ -plane both for RK2 and IEX4 stability domain.
- 3) Use IEX4 to simulate the transient response of the system with the parameters of point 2), and determine the largest step size such that RK2 yields a stable and accurate solution.
- 4) Discuss the results.

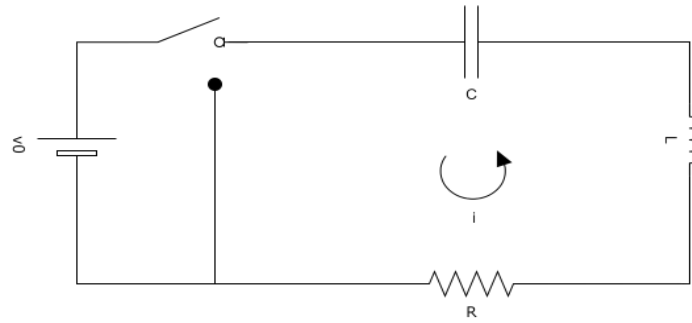


Figure 21: Electric circuit model.

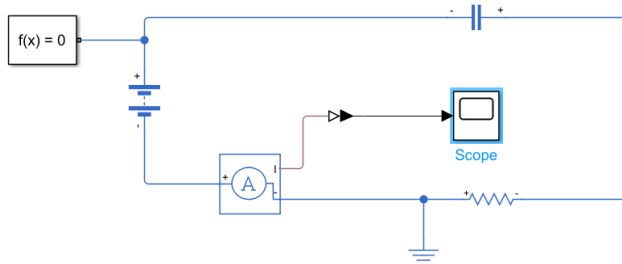
(4 points)

Solution

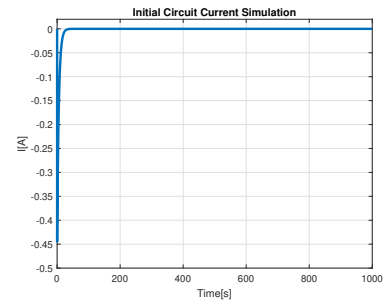
- 1) The state variable form of the system is shown in Eq. 22.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (22)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{1}{LC} & -\frac{R}{L} \end{bmatrix} \quad (23)$$

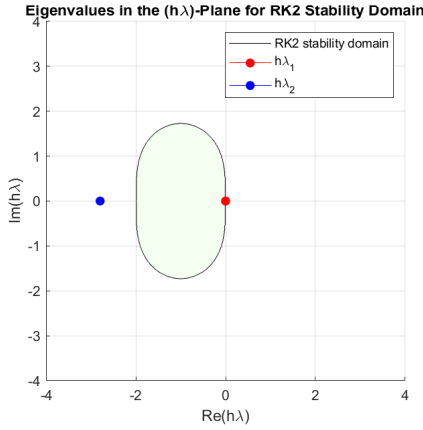


(a) Simulink circuit

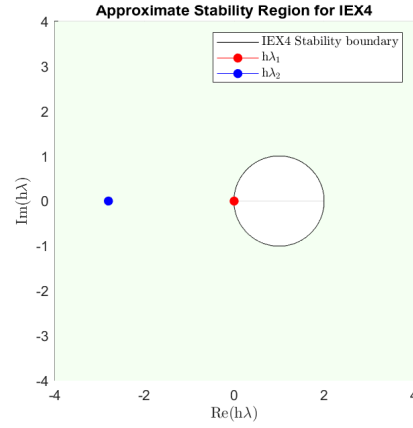


(b) Current solution for circuit simulation

Figure 22: Initial Values



(a) Eigenvalues in RK2 stability domain



(b) Eigenvalues in IEX4 stability domain

Figure 23: Stability Domains

2) Besides the given values, there were considered as initial values for the state variables $[V_0 \cdot C, I]$, where the current was considered to be $I = 0$, as it was obtained from the SIMULINK simulation of the system before the switch changes its position, Fig. 22. For the given parameters, in order to show that the system is stiff, the eigenvalues of matrix $\mathbf{A}$ (Eq. 23) were calculated, and it was considered stiff if $\lambda_2 < \lambda_1$, and if there was a significant difference in their order of magnitude. For this case, the system is considered stiff, if the magnitude difference between $\frac{\mathcal{O}(\lambda_2)}{\mathcal{O}(\lambda_1)} = 10^3$. Both conditions are satisfied and thus the system is stiff.

The eigenvalues were represented in the $(h\lambda)$ -plane assuming the maximum step size of RK4 method, $h_{max} = -\frac{2.8}{\lambda_2}$. This value was considered to show the difference between the RK2 and IEX4. The results are shown in Fig. 23a for RK2 and Fig. 23b for IEX4. It can be observed that for the IEX4 both eigenvalues belong to the stability domain, while for RK2 only the eigenvalue λ_1 corresponds to the stability domain. Thus, IEX4 is more suitable for stiff systems as the present one, than RK2, even if it is more computationally expensive.

3-4) The same initial values as in the previous case were considered for the state values, as the conditions in which the circuit works do not change. The transient response was calculated using the IEX4 method, considering a step size value of $h = 0.0002$, which would offer the desired tolerance, of 10^{-5} . The set of ten equations were solved applying Eq. 24, in which $\mathbf{k}$ represents the coefficients $(k_1, k_{2a}, k_2, k_{3a}, k_{3b}, k_3, k_{4a}, k_{4b}, k_{4c}, k_4)$, and $\mathbf{x}$ represents the initial condition of the equation, thus $\mathbf{x}(t)$ or the previous coefficient.

$$\mathbf{k} = (\mathbf{I}_2 - h\mathbf{A})^{-1}\mathbf{x}; \quad (24)$$

Calculating the coefficients in this way made the process less costly, compared to using *fsolve*. The transient response is shown in Fig. 24.

If a much smaller step size was chosen, the problem became more expensive, and the improvement in precision would not increase much, thus for these reasons the current step size was chosen.

The RK2 step size limit was determined using the *fsolve* on the function represented Eq. 25. This leads to the solution of the marginally stable value of h . The solution obtained is $h = 0.0016$, which is also obtained from $h = -2/\lambda_2$.

$$F(h) = \mathbf{I}_2 + h\mathbf{A} + \frac{h^2}{2}\mathbf{A}^2 \quad (25)$$

The transient response of the system is represented in Fig. 25 for different values of step-size. It is clear that for the first 3 cases the step-size is too large and thus the solution is not stable,

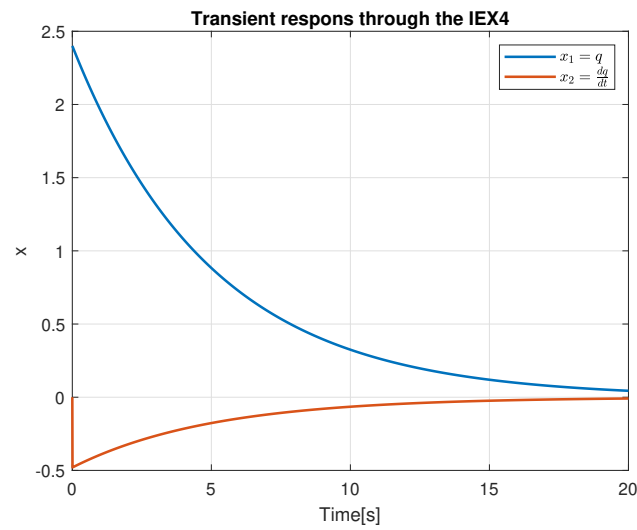


Figure 24: Transient Response

while for the last three the solution is stable. This includes the maximum step-size, h_{max} as this is the marginally stable solution.

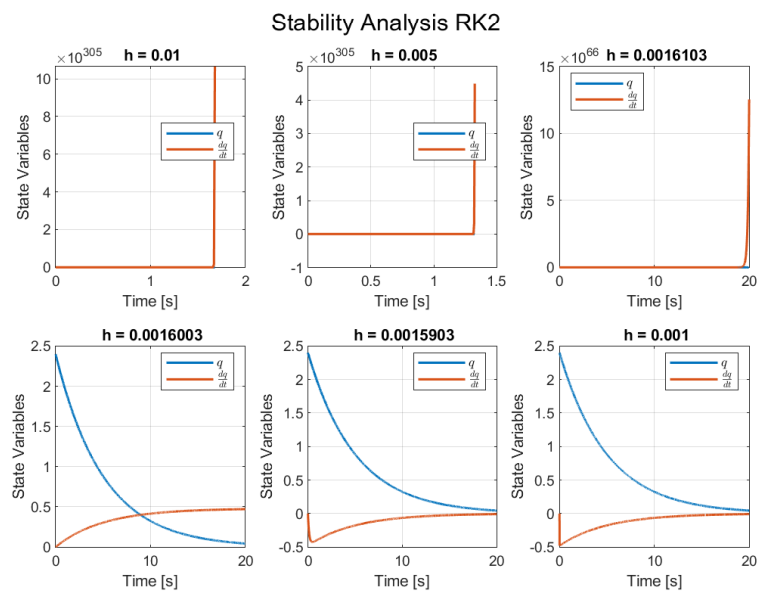


Figure 25: Transient Response for Different step sizes

Exercise 6

Consider the bouncing ball physical model in Fig. 26. The mathematical model of the system is represented by the following equations:

$$\frac{dv}{dt} = -g + \frac{F_D}{m_b} + \frac{F_A}{m_b} \quad (26)$$

$$\frac{dx}{dt} = v \quad (27)$$

$$v^+ = -k \cdot v^- \quad (28)$$

$$F_D = -0.5 \cdot \rho \cdot C_d \cdot A_b \cdot v \cdot |v| \quad (29)$$

$$F_A = \rho \cdot V_b \cdot g \quad (30)$$

where the initial conditions are: $v(0) = 0$ [m/s], and $x(0) = 10$ [m].

Assuming that: the ball velocity before (v^-) and after (v^+) the ground impact is governed by the attenuation factor $k = 0.9$, the air density is $\rho = 1.225$ [kg/m³], the ball mass is $m_b = 1$ [kg], the ball area is $A_b = 0.07$ [m²], the ball volume is $V_b = 0.014$ [m³], and the drag coefficient is $C_d = 1.17$, answer to the following tasks:

- 1) Solve the mathematical model in the time interval $t \in [0, 10]$ s using an ode integrator of Matlab.
- 2) Repeat point 1) for $\rho = 15$ [kg/m³]. Which is the difference with respect to point 1)?
- 3) Now solve the problem when the fluid density is $\rho = 60$ [kg/m³] in $t \in [0, 10]$ s. Discuss the result.
- 4) Repeat point 3) using RK4 for $h = 3$. Discuss the result.

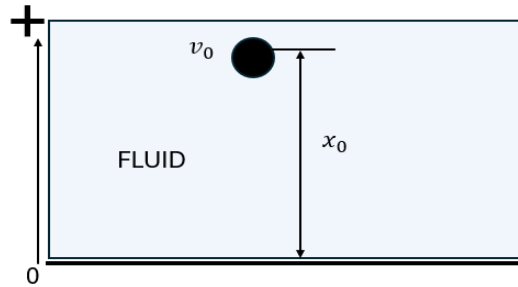


Figure 26: Bouncing ball system.

(5 points)

Solution

1) **Case 1** - $\rho = 1.225$ [kg/m³]

For the first part of the exercise, was used the *ode78*, as the system is neither stiff and nor linear. For the integration, the inputs were considered to be the time (**t**) and the ball's physical states, represented in vector form [**t**,**x**], and for all the following simulations, the initial conditions of the system were considered to be $x_0 = 10$ for the position and $v_0 = 0$ for the velocity. These were then introduced in the integrator for which the tolerances were chosen at 10^{-9} . It was chosen this value for the tolerance in order to consider the contact with the ground as rapidly as possible.

The contact with the ground was simulated through an *Event Function*, placed in the integrator, thus every time the ball simulation reaches the x-axis ($x = 0$), the simulation

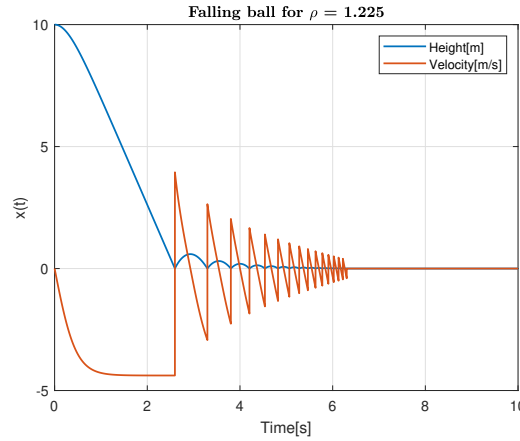


Figure 27: Falling ball for $\rho = 1.225$ with y-axis represented but the $x(t)$ state vector, composed of $[x(t) \ v(t)]$.

is stopped. This gives a specific time window in which the simulation ran ($[t_0 \ t_1]$). Afterwards, the value of velocity is updated considering the attenuation factor, and the simulation restarts from the last time step (t_1). The process is repeated until the ball reaches its equilibrium state, meaning that it doesn't bounce anymore. This state was considered considered from the results offered by the integrator. If the integrator reached its limit, the propagation would be performed in the negative part of the x-axis, which is undesirable, and thus for the last step in which this happens the values of x and v are set to zero for the remainder of time interval, in this case $[6.3, 10]$. The time - evolution of the ball's state is represented in Fig. 27. It must be noted that the sign of the velocity represents its direction, thus a negative velocity represents the ball falling, while a positive velocity represents the ball raising after hitting the ground.

2) Case 2 - $\rho = 15[kg/m^3]$

For the second case in which $\rho = 15$, it is observed the effect of this increase in evolution of the ball. As it can be seen in Fig. 28 that the ball touches the ground for the first time at a larger time instance with respect to the first case and the time interval is not large enough for the ball to reach equilibrium. The process of simulating the falling ball is similar to the previous case.

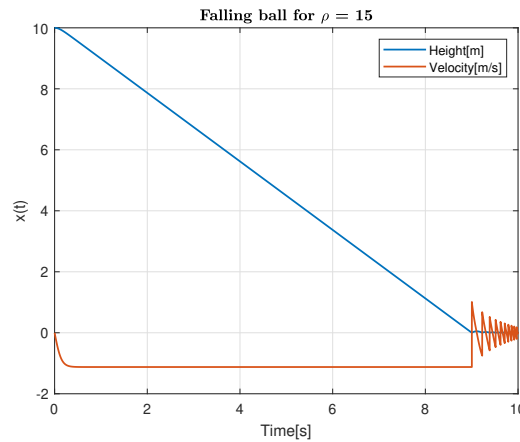


Figure 28: Falling ball for $\rho = 15$ with y-axis represented but the $x(t)$ state vector, composed of $[x(t) \ v(t)]$.

3) Case 3 - $\rho = 60[kg/m^3]$

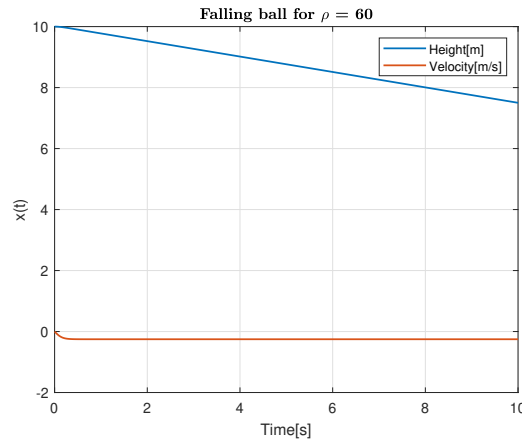


Figure 29: Falling ball for $\rho = 60$ with y-axis represented but the $x(t)$ state vector, composed of $[x(t) \ v(t)]$.

The density increase can be observed once again with respect to the other two cases. In this case the ball does not have the time to reach the ground and to bounce.

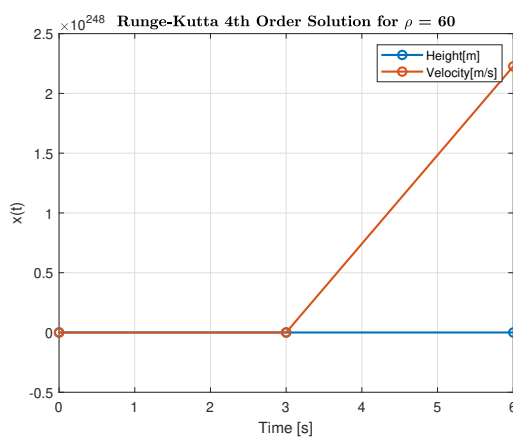
In Table 2 are given the times of each of the three cases for the first contact with the ground and if the ball reaches equilibrium.

$\rho \ [kg/m^3]$	First contact time[s]	Equilibrium (Y/N)
1.225	2.5961	Y
15	9.0032	N
60	-	N

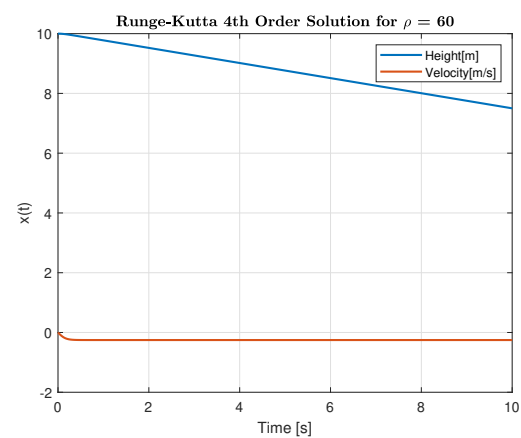
Table 2: Comparison of the three cases

4) Case 4 - RK4 method with $h = 3$

As in the previous exercises the RK4 method was implemented dividing the time interval by considering the possible number of steps, thus, $\text{round}(\frac{\Delta t}{h})$. However, as it can be seen in Fig. 30a, an $h = 3$ is too large for this problem, and the RK4 model cannot solve the problem.



(a) Solution for $h = 3$



(b) Solution for $h = 0.03$

Figure 30: RK4 method for $\rho = 60$

This is understandable as in RK4 large step sizes can introduce instability, and thus the solution diverges, from the analytical solution. This can be observed in Fig. 30a,

where the solution reaches infinity. However, if the step size is updated, such that a new optimal time step is reached, the RK4 model offers the same solution with the analytical one, which can be seen in Fig. 30b.

The stability was found by changing h until it reaches 0.03. For this it was calculated the error between RK4 and RK5, and the tolerance for this was set equal to 10^{-6} . Finally, in Fig. 31 is shown the error between the analytical and RK4 solutions. This is acceptable as for this case, the integrator tolerance was set to 10^{-3} , as the ball did not encounter the ground.

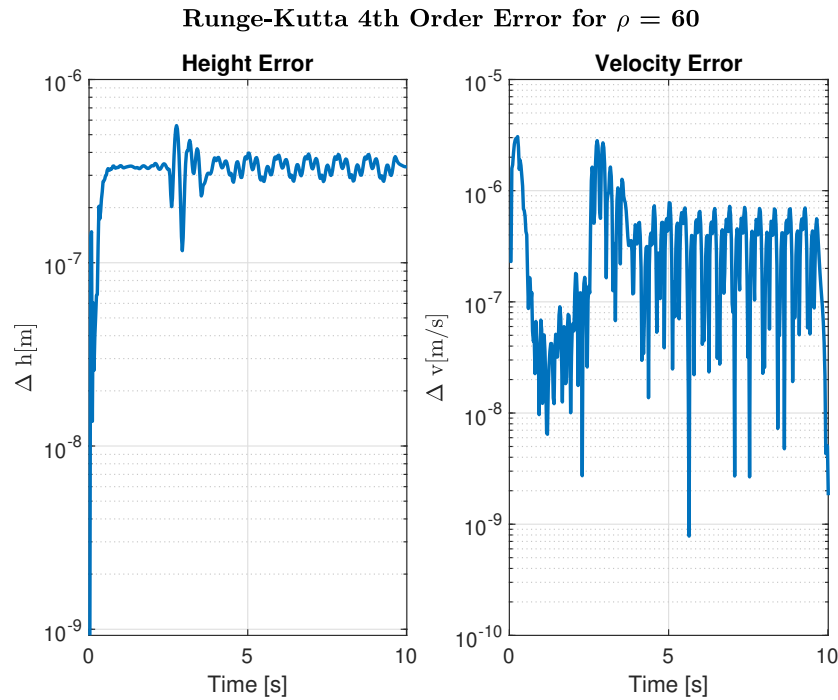


Figure 31: Error between RK4 and analytical for $\rho = 60$